

Stator-Flux Oriented Control of the Induction Machine

A Step-by-Step Derivation

1 Motivation

In indirect rotor-flux oriented control, the controller is built from a reference i_{ds} , a reference i_{qs} , and a current regulator. The torque demand sets i_{qs} directly, and the desired rotor flux level sets i_{ds} . However, locating the rotating reference frame requires the rotor flux angle, and that angle is normally reconstructed from *reference* currents rather than measured ones. Two questions remain open:

- (i) How do we find the frame angle θ needed for the reference-frame transformation?
- (ii) How do we determine a proper i_{ds}^* value, i.e. the one that maintains a desired constant flux level in the machine?

Stator-flux oriented control answers both questions by orienting the rotating frame on the *stator* flux instead of the rotor flux.

2 Why the Stator Flux?

The stator flux linkage is the one most easily estimated from terminal measurements:

$$\hat{\bar{\lambda}}_s = \int (\bar{v}_s - r_s \bar{i}_s) dt \quad (1)$$

In practice, pure integration of the back-EMF drifts: the estimated flux magnitude tends to ramp away from the true value over time, rather than tracking it. This is the same drift problem encountered with rotor-flux observers, and it is removed the same way – e.g. with a closed-loop flux observer (CLFO) that corrects the integrator drift.

3 Governing Machine Equations

Stator side

$$\bar{v}_s = r_s \bar{i}_s + p \bar{\lambda}_s + j\omega_e \bar{\lambda}_s, \quad \bar{\lambda}_s = L_s \bar{i}_s + L_m \bar{i}_r \quad (2)$$

where $p = d/dt$, ω_e is the rotating speed of the chosen dq frame, and $L_s = l_l + L_m$.

Rotor side

$$\bar{v}_r = r_r \bar{i}_r + p \bar{\lambda}_r + j(\omega_e - \omega_r) \bar{\lambda}_r = 0, \quad \bar{\lambda}_r = L_r \bar{i}_r + L_m \bar{i}_s \quad (3)$$

where ω_r is the rotor shaft speed in electrical rad/s, $L_r = l_{lr} + L_m$, and $\bar{v}_r = 0$ because the rotor windings are short-circuited.

Torque equation

$$\tau = \frac{3}{2}\hat{p}\text{Im}\left(\bar{i}_{dqs} \cdot \bar{\lambda}_{dqs}^*\right) \longrightarrow \tau = \frac{3}{2}\hat{p}\lambda_{ds}i_{qs} \quad (4)$$

($\hat{p}$ here denotes the number of pole pairs, $\text{Im}(\cdot)$ the imaginary part.)

4 Choosing the Torque Equation and the Orientation

A general principle applies regardless of which flux is used for orientation: the stator currents always appear in the torque equation, since they are the quantities we can measure and control directly.

- Rotor-flux oriented control $\Rightarrow$ torque equation uses *rotor flux* and stator currents.
- Stator-flux oriented control $\Rightarrow$ torque equation uses *stator flux* and stator currents.

For stator-flux orientation, the torque is

$$T = \frac{3}{2}\hat{p}\text{Imag}\left([i_{ds} + ji_{qs}] \cdot [\lambda_{ds} - j\lambda_{qs}]\right) = \frac{3}{2}\hat{p}(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}) \quad (5)$$

Now the d -axis of the rotating frame is aligned with $\bar{\lambda}_s$ itself, at angle θ_s from the stationary a -axis, rotating at $\omega_e = 2\pi f$. By construction of this frame,

$$|\bar{\lambda}_{dqs}| = \lambda_{ds}, \quad \boxed{\lambda_{qs} \equiv 0} \quad (6)$$

This is the key simplifying condition used throughout the rest of the derivation. With $\lambda_{qs} = 0$, the torque reduces to

$$T = \frac{3}{2}\hat{p}\lambda_{ds}i_{qs} \quad (7)$$

which has exactly the same form as a PMSM torque equation, $T = \frac{3}{2}\hat{p}\lambda_{pm}i_{qs}$, with λ_{ds} playing the role of the (controllable) magnet flux.

Two tasks remain:

- ① θ_s is found as $\angle \bar{\lambda}_s$, estimated from $\int(\bar{v}_s - R_s\bar{i}_s)dt$ (Section 2).
- ② How do we maintain a constant λ_{ds} ? $\rightarrow$ we must go back to the machine equations.

5 Eliminating the Rotor Current (Principle 1)

Start from the two flux-linkage equations:

$$\bar{\lambda}_s = L_s\bar{i}_s + L_m\bar{i}_r, \quad \bar{\lambda}_r = L_r\bar{i}_r + L_m\bar{i}_s \quad (8)$$

Solve the rotor equation for $\bar{i}_r$:

$$\bar{i}_r = \frac{\bar{\lambda}_r - L_m\bar{i}_s}{L_r} \quad (9)$$

Substitute into the stator flux equation:

$$\bar{\lambda}_s = L_s\bar{i}_s + L_m\frac{\bar{\lambda}_r - L_m\bar{i}_s}{L_r} = \left(L_s - \frac{L_m^2}{L_r}\right)\bar{i}_s + \frac{L_m}{L_r}\bar{\lambda}_r \quad (10)$$

Introduce the leakage factor $\sigma = \frac{L_s L_r - L_m^2}{L_s L_r}$ (small, since $L_s = l_{ls} + L_m > L_m$ and $L_r = l_{lr} + L_m > L_m$; $\sigma \rightarrow 0$ as leakage $\rightarrow 0$):

$$\bar{\lambda}_s = \sigma L_s \bar{i}_s + \frac{L_m}{L_r} \bar{\lambda}_r \quad (11)$$

Therefore

$$\frac{L_m}{L_r} \bar{\lambda}_r = \bar{\lambda}_s - \sigma L_s \bar{i}_s \implies \bar{\lambda}_r = \frac{L_r}{L_m} \bar{\lambda}_s - \sigma \frac{L_s L_r}{L_m} \bar{i}_s \quad (12)$$

This relation alone already lets the rotor flux be estimated purely from stator-side quantities – the basis of (direct) rotor-flux oriented sensorless control. Here it is used as an intermediate step toward stator-flux orientation.

6 Eliminating the Rotor Current and Rotor Flux from the Rotor Voltage Equation (Principle 2)

The rotor voltage equation is

$$\bar{v}_r = r_r \bar{i}_r + (p + j s \omega_e) \bar{\lambda}_r \quad (13)$$

(stator currents are always retained, since they are measurable and controllable). Substitute $\bar{i}_r$ and $\bar{\lambda}_r$ from (12). Using the identity

$$\frac{L_m}{L_r} (\bar{\lambda}_r - L_m \bar{i}_s) = \bar{\lambda}_s - L_s \bar{i}_s$$

(a direct rearrangement of the stator flux-linkage definition) and defining the *rotor time constant*

$$\tau_r = \frac{L_r}{r_r}, \quad (14)$$

the rotor voltage equation collapses, after setting $\bar{v}_r = 0$, to a single equation containing only the stator flux, the stator current, and machine parameters:

$$\boxed{0 = (\bar{\lambda}_s - L_s \bar{i}_s) + \tau_r (p + j s \omega_e) (\bar{\lambda}_s - \sigma L_s \bar{i}_s)} \quad (15)$$

This is the master equation of stator-flux oriented control: every rotor-side variable has been eliminated.

7 Decomposition into d, q Components and the Orientation Simplification (Principle 3)

Writing $\bar{\lambda}_s = \lambda_{ds} + j \lambda_{qs}$ and combining coefficients, equation (15) becomes

$$0 = (1 + \tau_r p + j s \omega_e \tau_r) \bar{\lambda}_s - (L_s + \sigma L_s \tau_r p + j s \omega_e \sigma L_s \tau_r) \bar{i}_s \quad (16)$$

which splits into real (d) and imaginary (q) parts:

$$0 = (1 + \tau_r p) \lambda_{ds} - s \omega_e \tau_r \lambda_{qs} - (1 + \sigma \tau_r p) L_s i_{ds} + s \omega_e \sigma L_s \tau_r i_{qs} \quad (17)$$

$$0 = (1 + \tau_r p) \lambda_{qs} + s \omega_e \tau_r \lambda_{ds} - (1 + \sigma \tau_r p) L_s i_{qs} - s \omega_e \sigma L_s \tau_r i_{ds} \quad (18)$$

Now apply the orientation condition from Section 4: $\lambda_{qs} = 0$. Every term containing λ_{qs} vanishes, leaving two compact equations that form the core of the controller:

$$0 = (1 + \tau_r p) \lambda_{ds} - (1 + \sigma \tau_r p) L_s i_{ds} + s \omega_e \sigma L_s \tau_r i_{qs} \quad (1)$$

$$0 = s \omega_e \tau_r \lambda_{ds} - (1 + \sigma \tau_r p) L_s i_{qs} - s \omega_e \sigma L_s \tau_r i_{ds} \quad (2)$$

8 Slip Estimation from Equation (2) (Principle 4)

Solving (2) for $s\omega_e$:

$$s\omega_e = \frac{(1 + \sigma\tau_r p)L_s i_{qs}}{\tau_r \lambda_{ds} - \sigma L_s \tau_r i_{ds}} \quad (19)$$

where i_{ds}, i_{qs} come from measurement and λ_{ds} comes from the back-EMF integration (Section 2). Rearranging into a more compact form:

$$s\omega_e = \frac{\left(\frac{1}{\sigma\tau_r} + p\right) i_{qs}}{\frac{\lambda_{ds}}{\sigma L_s} - i_{ds}} \quad (3)$$

This gives the slip frequency needed to correctly locate the rotating frame – answering question (i) from Section 1, but now using the estimated stator flux rather than reference currents.

9 The i_{ds} Reference from Equation (1) (Principle 5)

The reference i_{qs}^* is set directly by the torque command, since $T = \frac{3}{2}\hat{p}\lambda_{ds}i_{qs}$. Equation (1) rearranges into the companion relation that gives the i_{ds} reference needed to hold λ_{ds} at its desired level:

$$\left(\frac{1}{\sigma\tau_r} + p\right) i_{ds} = \left(\frac{1}{\tau_r} + p\right) \frac{\lambda_{ds}}{L_s} + s\omega_e i_{qs} \quad (4)$$

This answers question (ii) from Section 1: how to choose i_{ds}^* to maintain the desired flux level.

10 Assembling the Control Structure

Equations (3) and (4) together define the controller:

- A stator-flux estimator $\bar{\lambda}_s = \int(\bar{v}_s - R\bar{i}_s) dt$ provides $\hat{\bar{\lambda}}_s$. Its magnitude is compared against λ_s^* in a flux PI controller; the PI output $\Delta\lambda$ corrects for the parameter errors that the open-loop relation (4) would otherwise accumulate.
- λ_s^* is passed through $1/(\sigma L_s)$ and a $(p + 1/\tau_r)$ block, summed with $\Delta\lambda$, then passed through $1/(p + 1/\sigma\tau_r)$ to produce i_{ds}^* – i.e. equation (4) implemented as a transfer-function block, with the PI loop closing the parameter-mismatch gap.
- A speed controller produces the torque command, which is scaled (via $2/(3\hat{p})$ and division by λ_{ds}) into i_{qs}^* . Together with λ_{ds} and i_{ds} , this drives the slip calculation of equation (3), whose output is integrated ($1/s$) to obtain the flux angle γ used in the frame-rotation transformation.

11 A Simpler Alternative Implementation

Starting again from (4) and solving directly for i_{ds} :

$$i_{ds} = \frac{(1 + \tau_r p)}{(1 + \sigma\tau_r p)} \cdot \frac{\lambda_{ds}}{L_s} + \frac{s\omega_e \sigma\tau_r i_{qs}}{(1 + \sigma\tau_r p)} \quad (20)$$

Since in the stator-flux oriented frame $\lambda_{ds} = \lambda_s$, the first term can be replaced directly by a PI controller acting on the flux error, rather than implementing the lead-lag transfer function explicitly:

$$i_{ds}^* = G_{PI}(\lambda_s^* - \hat{\lambda}_s) + \frac{s\omega_e i_{qs}}{\left(\frac{1}{\sigma\tau_r} + p\right)} \quad (21)$$

In steady state, $p = d/dt = 0$, so the second term simplifies further:

$$\boxed{i_{ds}^* = G_{PI}(\lambda_s^* - \hat{\lambda}_s) + s\omega_e \sigma\tau_r i_{qs}} \quad (22)$$

This gives a much simpler practical realization than the full block diagram of Section 8: a flux PI loop plus a feed-forward decoupling term, instead of the explicit lead-lag transfer-function path.

Summary of the Derivation

1. Estimate $\bar{\lambda}_s$ from voltage/current via back-EMF integration (with drift correction).
2. Orient the rotating frame on $\bar{\lambda}_s$, forcing $\lambda_{qs} = 0$.
3. Eliminate the rotor current using the flux-linkage equations (Principle 1).
4. Eliminate the rotor flux and rotor current from the rotor voltage equation, leaving an equation in $\bar{\lambda}_s, \bar{i}_s$ only (Principle 2).
5. Decompose into d, q and apply $\lambda_{qs} = 0$ to get two compact equations (Principle 3).
6. One equation gives the slip estimate $s\omega_e$, used to build the frame angle (Principle 4).
7. The other equation gives the i_{ds}^* reference needed to hold the flux constant (Principle 5).
8. Assemble both into the flux/torque control structure, optionally simplified with a PI loop in steady state.